

Resilient Humanoid Loco-Manipulation

Learned whole-body policies and online layer adaptation under actuator faults

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8.7 s

whole-body box pickup

57

logged hardware runs

+53%

survival under a knee fault

50 Hz

online adaptation, numpy

A policy is only correct for the robot it was trained on

- **Every policy is optimized against one fixed set of dynamics:** nominal PD gains, a known payload, a particular contact model.
- **Real robots leave that set.** Actuators weaken, payloads change, and the sensors the policy trained on are not the sensors the hardware has.
- **A frozen policy cannot notice.** It keeps issuing the commands that were optimal for a robot it is no longer controlling.

Objective

Behaviors that survive conditions outside the training distribution, and a controller that responds to changes in the robot's own dynamics without being told what changed.

OBSERVED ON THIS ROBOT

Feet on their edges

Sim reported contact force, so the reward counted the foot as planted.

Saturated ankle roll

Frontal-plane correction ran through a joint with 15° of travel.

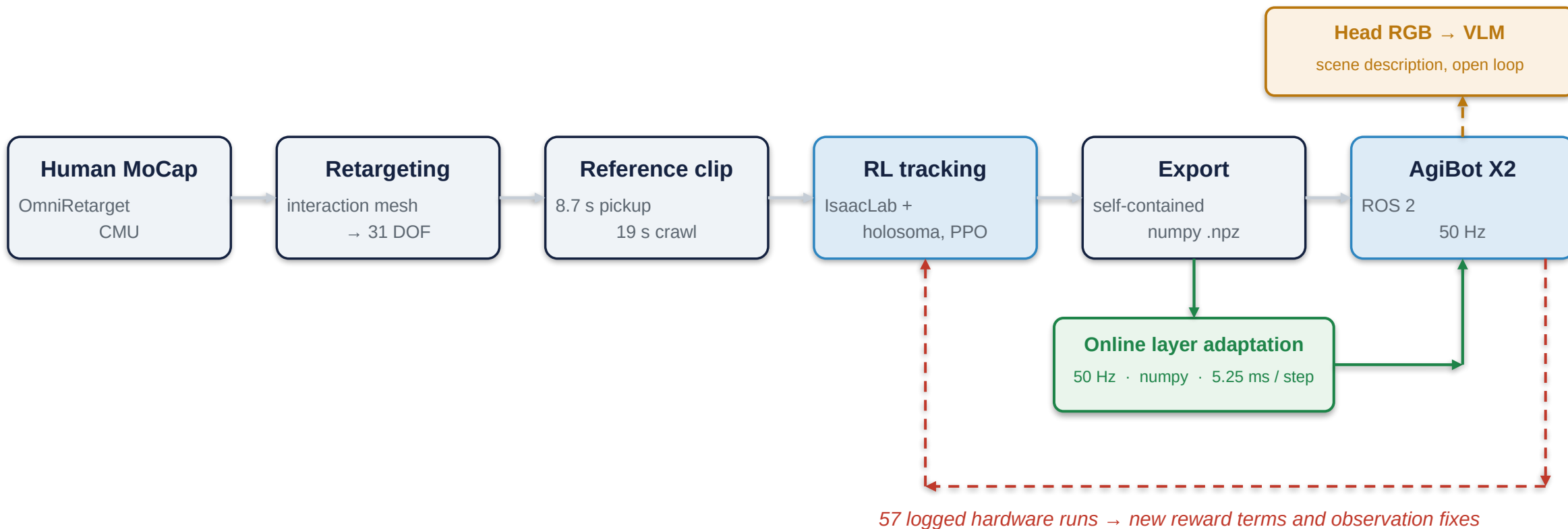
Wrong base frame

Torso IMU fed into an observation holosoma builds from the pelvis.

Weakened actuator

Right knee at 30% stiffness. Survival falls from 14.7 s to 7.4 s.

From human motion capture to a policy on the robot



4096 parallel envs 200 Hz physics / 50 Hz control blind actor MLP 512-256-128 (ELU) DR: box mass, friction, PD gain, action delay, encoder noise, pushes

Box pickup: one policy for the whole 8.7 second motion

From the default standing pose the robot bends to a 45 cm box, squeezes it between two rigid fingerless palms, lifts it to chest height, carries it 1.4 m, and sets it down over planted feet.

No fingers, no cameras, and the payload moves the centre of mass while the robot is bent over.

VIDEO 1 Early · v8, iter 8000

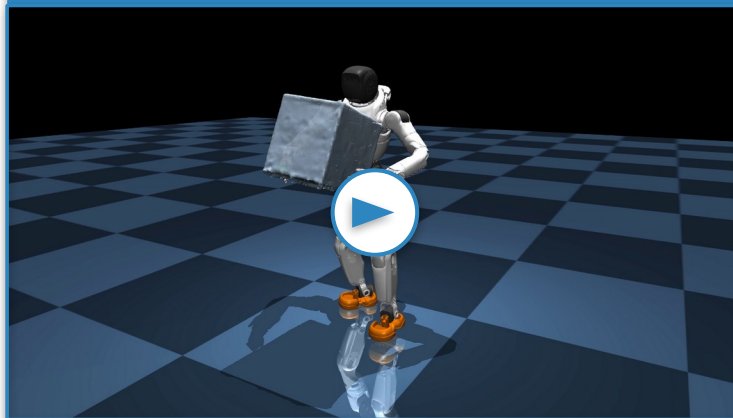


Balances through the full motion but pantomimes the lift. The box never leaves the floor.

x2_box_v8_iter8000_full.mp4

One continuous reference motion, no state machine

VIDEO 2 Intermediate · v10, iter 28000



First complete pickup, carry and set-down in simulation after the two-hand grasp reward.

x2_box_v10_iter28000_final.mp4

VIDEO 3 Current · v33, iter 253000



Flat planted feet, quiet waist, returns fully upright. This is the deployed policy.

x2_box_v33_waist_track_iter253000.mp4

hold upright

bend down

grasp +
lift

carry 1.4 m

set down

stand up

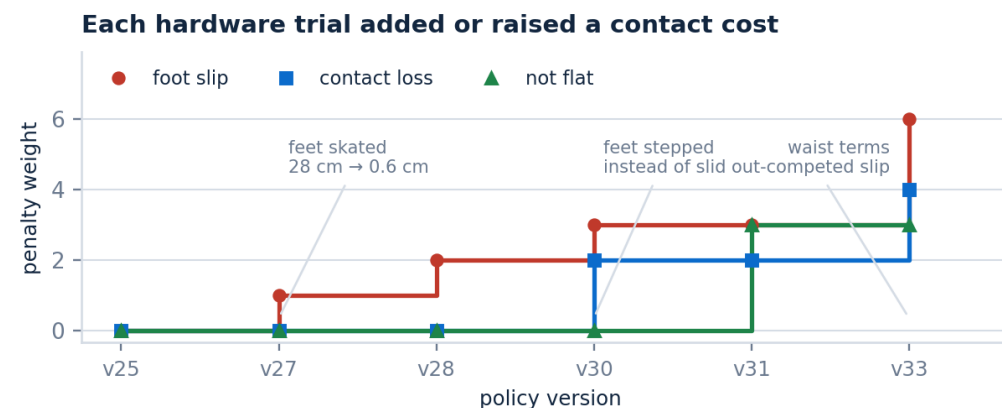
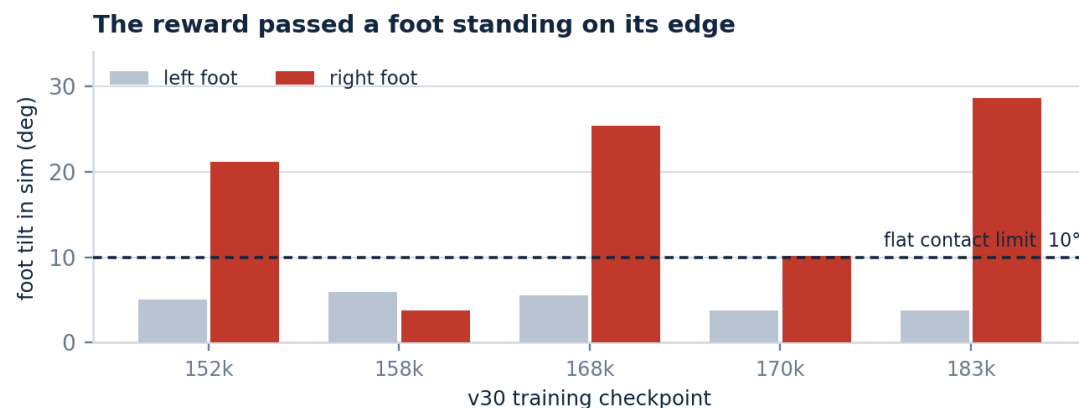
0 s

8.7 s

Most of the work was finding out why it did not work

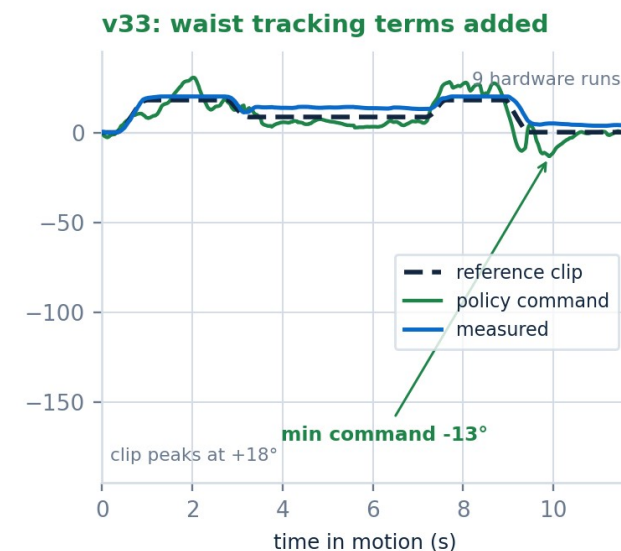
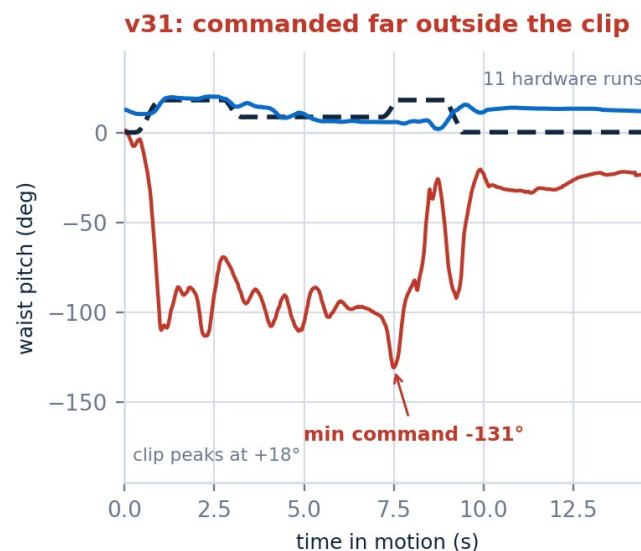
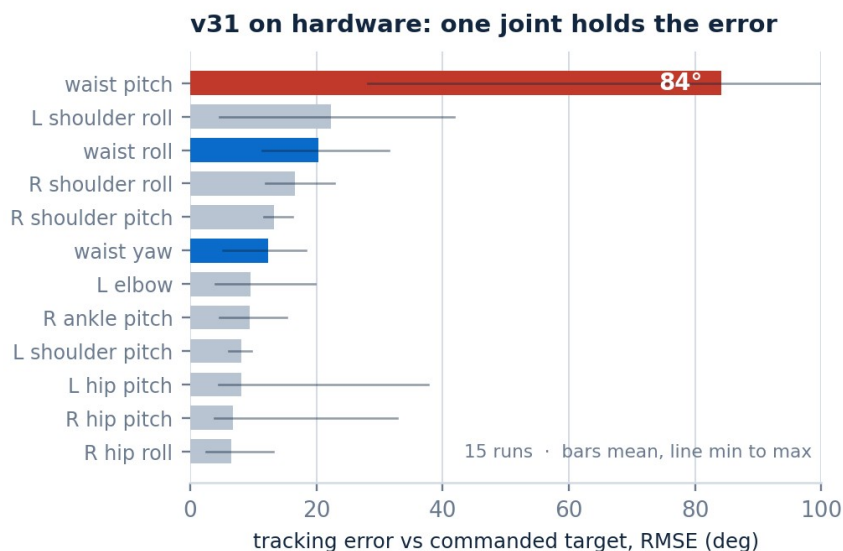
v8	Forgot the walk-up	Sampler mixed raw failure counts, so the uniform term was a no-op.	Mix normalized probabilities.
v10	Pantomimed the lift	A mean over hands let one palm mask the other hovering away.	Grasp reward as a product over both hands.
v13	Frozen hip, impossible waist	Off-by-one in the retargeter clamped every joint to the previous joint's limits.	Index limits by joint type, retrain.
v24	Crab-walked the carry	The demonstrator turned 65 degrees and shuffled his feet 30 cm.	Straighten the carry, pin the feet.

Then hardware found what simulation could not. A foot on its edge still reports contact force, and a tilted stationary foot has zero slip.



The hardware logs pointed at a single joint

Version 31 was the best policy in simulation and it fell on the robot, 13 of 15 runs, always at the same point in the motion.



SYMPTOM

Torso pitched forward and collapsed at $t \approx 8.2$ s, entering the carry. Leg tracking error stayed near 5° throughout.

CAUSE

The actor drove the waist pitch command to -131° while the reference clip asks for at most $+18^\circ$. Nothing in the reward priced the waist.

FIX, V33

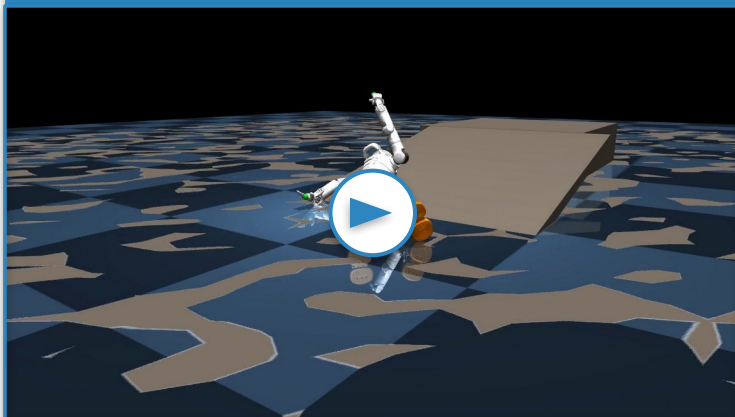
Waist pitch and yaw tracking terms, exponential plus L2. Command back inside the clip and hardware waist error 13.1° to 3.3°.

Crawling: a prone gait needs its own reward set

A whole-body prone crawl up a slope, retargeted from a CMU motion capture clip and trained in the same holosoma stack. The robot moves on palms, knees and feet.

The upright reward suite is hostile to crawling. Foot slip penalizes the limb sliding a crawl needs, and contact loss demands both feet stay loaded. The task needs its own reward, contact and observation set.

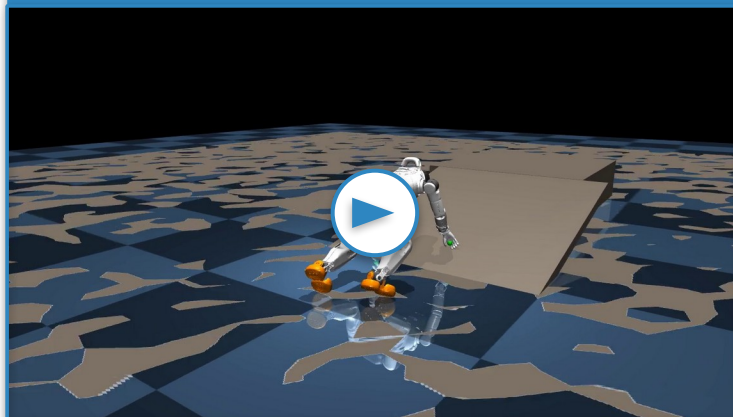
VIDEO 1 Early · v2, iter 14500



Crawls for the first time. Tracking is loose and the robot drifts off the slope.

x2_crawl_slope_v2_palmflat_iter14500.mp4

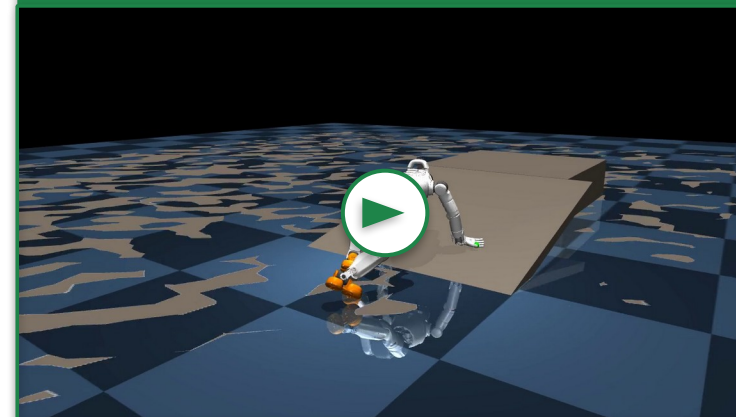
VIDEO 2 Intermediate · v3, iter 49999



Retuned tracking weights. This checkpoint was exported and run on the physical X2.

x2_crawl_slope_v3_tracking_iter49999.mp4

VIDEO 3 Current · v5, iter 86000



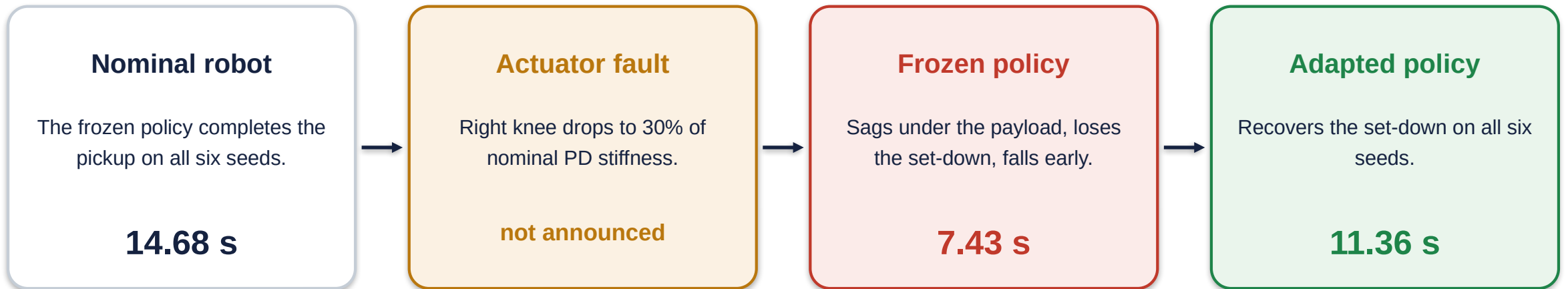
Position weight raised to 3.0. Mean episode 7.6 s, the best climb so far in simulation.

x2_crawl_slope_v5_track_xyz_iter86000.mp4

Honest limitations. v4 over-smoothed the gait until it could not generate thrust, and v6 multi-clip stand-up training regressed the climb from 24.9 to 16.2 mean reward. Best episode is still 7.6 s of a 19 s clip.

A good policy is still a fixed policy

To test that, the right knee actuator is scaled to 30% of nominal PD stiffness. It still moves, but it sags under load.

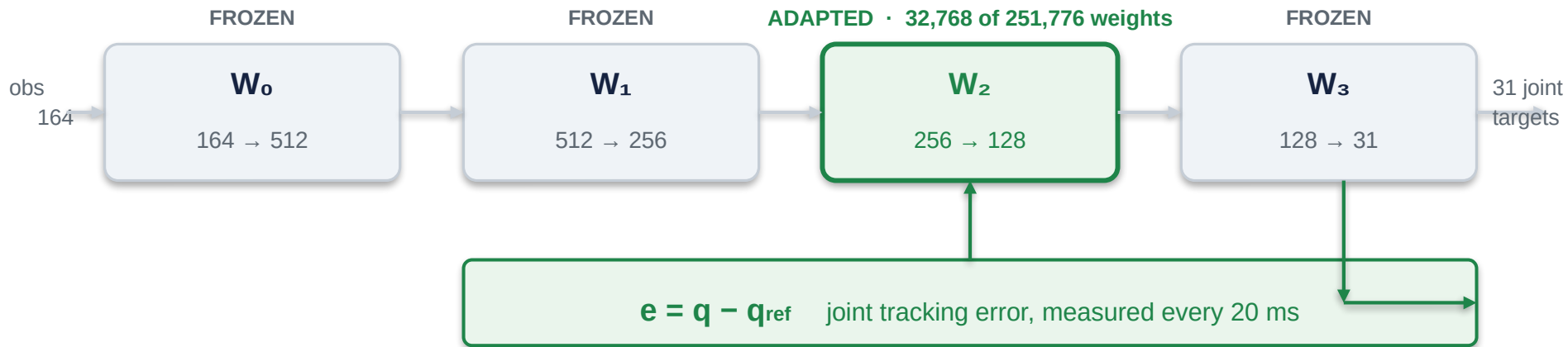


The fault is never announced. Neither the policy nor the adapter is told that a fault occurred. Both keep using the nominal gains. There is no detector and no trigger.

This is the regime the method is actually for. The same adaptation applied to a healthy robot spends balance margin the task needed and makes things worse. That measurement is in backup.

Adapt one layer, online, from tracking error alone

Ported from Taheri, Chung and Hadaegh, ACC 2026, “Closing the Loop Inside Neural Networks: Causality-Guided Layer Adaptation for Fault Recovery Control” — moved from its MuJoCo testbed into our IsaacLab environment on the v31 policy.



$$\dot{W} = \Gamma \delta z^T - \gamma (W - W_0)$$

Lyapunov update, leakage toward the trained weights

Adapted one weight matrix, 13% of the network

Frozen every other layer, the reward, the reference clip

Needs no plant model, no fault detector, no offline phase

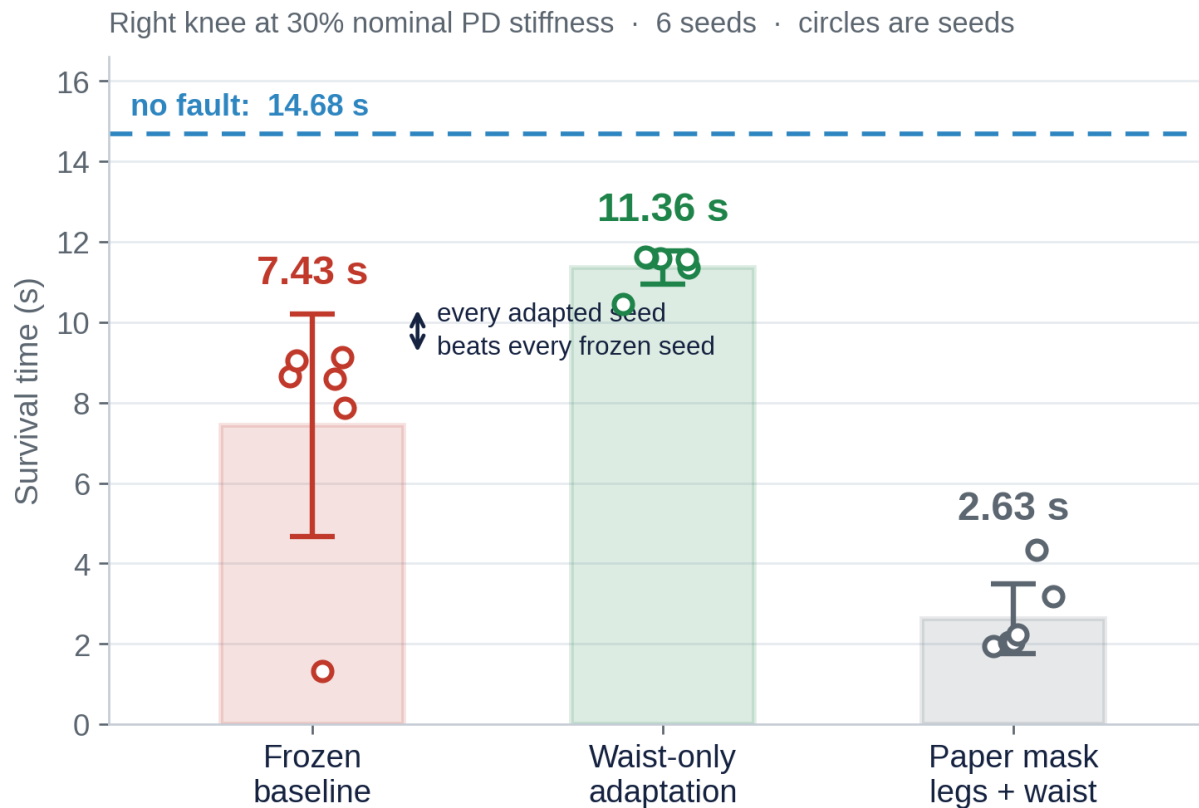
Cost 1.93 ms forward + 3.31 ms update = 26% of the 20 ms budget

Mask which joints enter e . Waist only, against the paper's hips, knees, ankles and waist

Safety resets to W_0 every episode; a norm bound trips a hard revert

Defect found and fixed: the shipped leak pulls the layer toward the origin rather than the trained weights, eroding 13.7% of its norm per episode even with perfect tracking. Fixing it does not change the outcome, but it is a real bug.

Under the fault, adaptation recovers the task



	Survival	Set-down	Lifted
Frozen baseline	7.43 ± 2.76 s	4 / 6	5 / 6
Waist-only adaptation	11.36 ± 0.42 s	6 / 6	6 / 6

6 seeds · no domain randomization · observation noise off · v31 policy

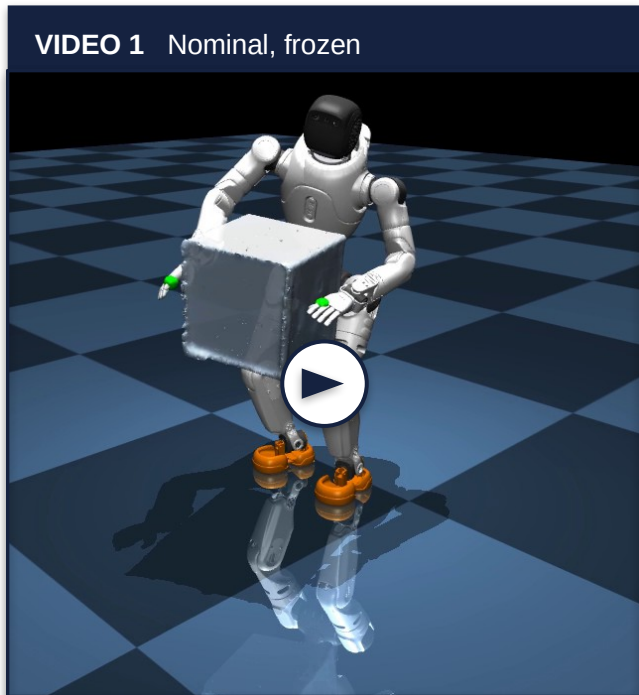
- **Complete separation.** Every adapted seed outlasts every frozen seed. Exact one-sided Mann-Whitney $p = 0.0011$, the floor for six against six.
- **Consistency, not just mean.** Spread collapses from ± 2.76 s to ± 0.42 s. The seed where the frozen policy crumples at 1.3 s is carried to 10.4 s.
- **The task comes back.** Frozen reaches the set-down on 4 of 6. Adaptation reaches it on all 6.

THE MASK DECIDES, NOT THE GAIN

Waist-only runs the same $3e-4$ gain that is catastrophic with the paper's legs-and-waist mask: 2.63 s and never lifts the box. The legs are where balance lives on a floating base, so regulating their tracking error fights the balance the policy is maintaining.

Same policy, same fault, one difference

All three clips run the v31 box-pickup policy. The only change between the middle and the right is whether the adaptation update is switched on.



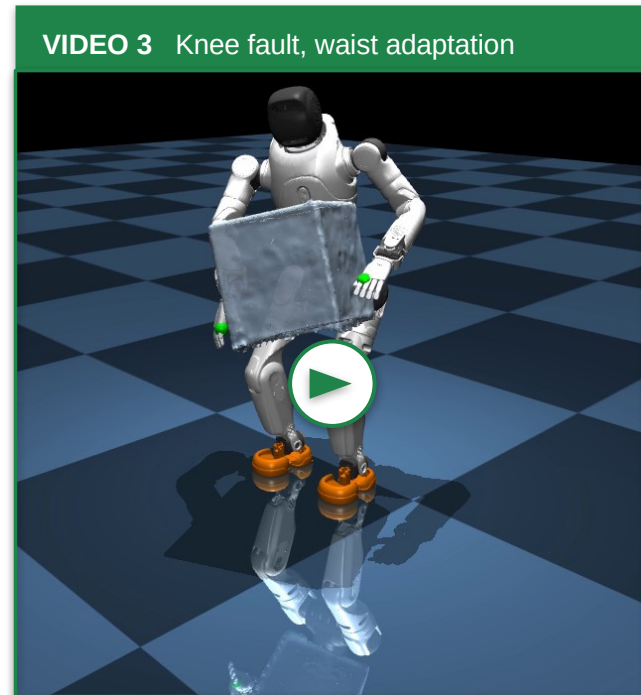
Healthy robot. Completes the pickup, carry and set-down. 14.68 s, 6/6 seeds.

x2_box_v31_flatfoot_iter202500.mp4



Right knee at 30%. Sags under the payload and falls before the set-down. 7.43 s.

isaac_fault_knee03_frozen_vs_waistadapt.mp4
(left half)

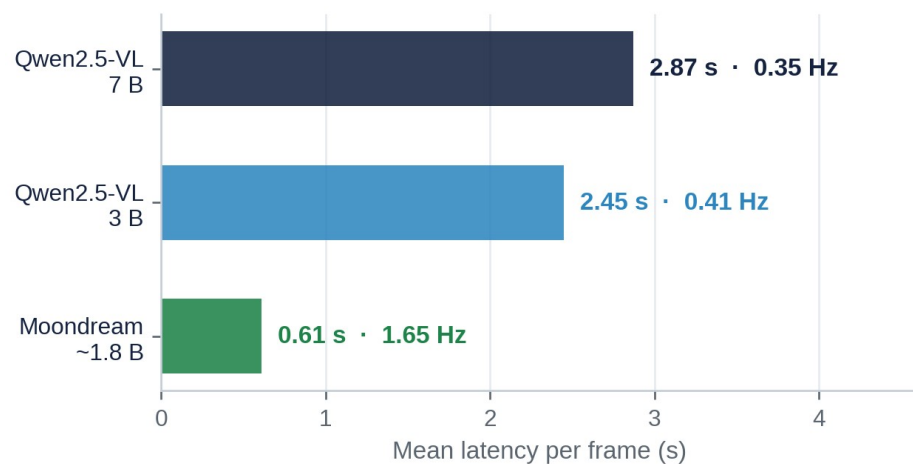
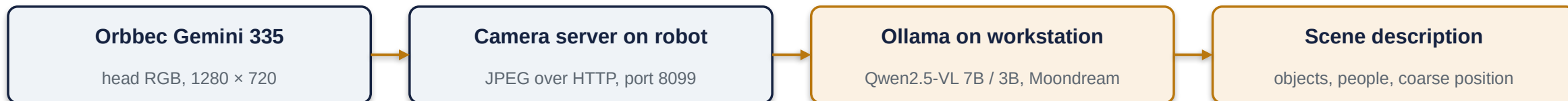


Same fault, adaptation on. Holds the box and reaches the set-down. 11.36 s.

isaac_fault_knee03_frozen_vs_waistadapt.mp4
(right half)

VLM perception: built and benchmarked, not yet closed on control

Adaptation handles faults the robot can feel in its own tracking error. Deciding what to do next needs the robot to understand the scene, which is where a vision language model fits.



Warm, end to end, RTX 5070 8 GB. 12 frames from one 15 s clip, identical prompt, temperature 0.2.

COMPLETED

- Head camera streamed off the robot
- Three models served locally through Ollama
- Latency and description quality measured

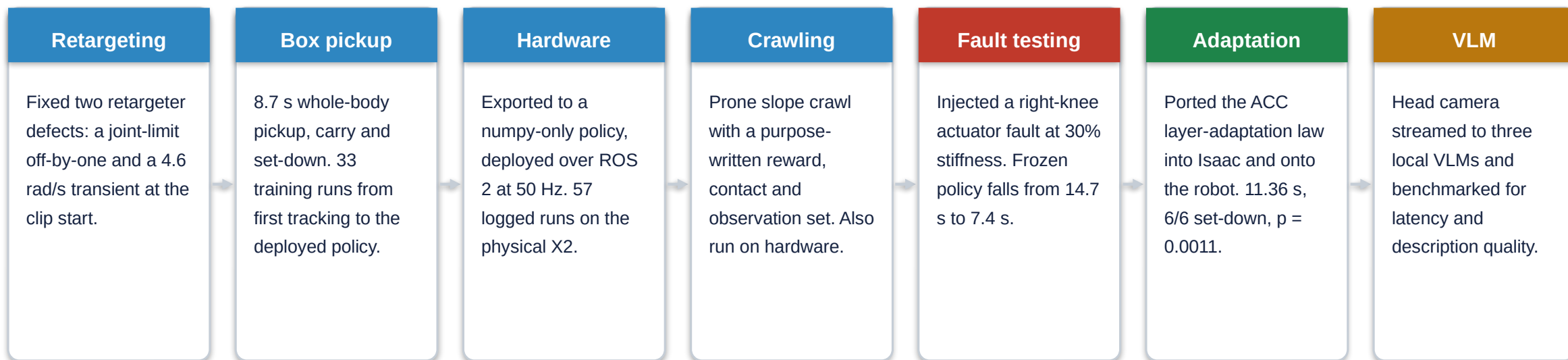
CURRENTLY TESTING

- Which model is accurate enough at a usable rate

NEXT STEP

- Free text into a structured signal the controller can consume

What this summer produced



Contributed here: the X2 integration and retargeting fixes, task and reward design in holosoma, every training run, the numpy export and ROS 2 deployment, the fault injection harness, and the adaptation port in both simulation and hardware form.

Built on holosoma for whole-body tracking, the OmniRetarget and CMU motion datasets, and the ACC 2026 adaptation law.

Where the system stands, and what comes next

WHAT THE SYSTEM DOES TODAY

Box pickup

8.7 s whole-body motion in simulation and on the physical X2, with flat planted feet and a clean set-down.

Crawling

Prone slope crawl trained in simulation, v3 exported and run on hardware.

Deployment

Numpy-only inference at 50 Hz over ROS 2, with a safety ladder and per-tick logging.

Adaptation

Runs online at 50 Hz in simulation and in the hardware deploy loop, and recovers the task under a knee fault.

NEXT STEPS, IN PRIORITY ORDER

1. Generalize the fault result

A hip or ankle fault, and a held-out seed pool.

2. Run the comparison on hardware

Both arms already run on the robot. What is missing is degrading a physical joint.

3. A balance-targeted error signal

Regulating joint error assumes balance follows. Define the error on a centre-of-mass term instead.

4. Gate the adaptation

Engage above an error threshold to restore a no-worse-than-frozen floor.

5. Close the perception loop

VLM output as a structured signal the controller can act on.

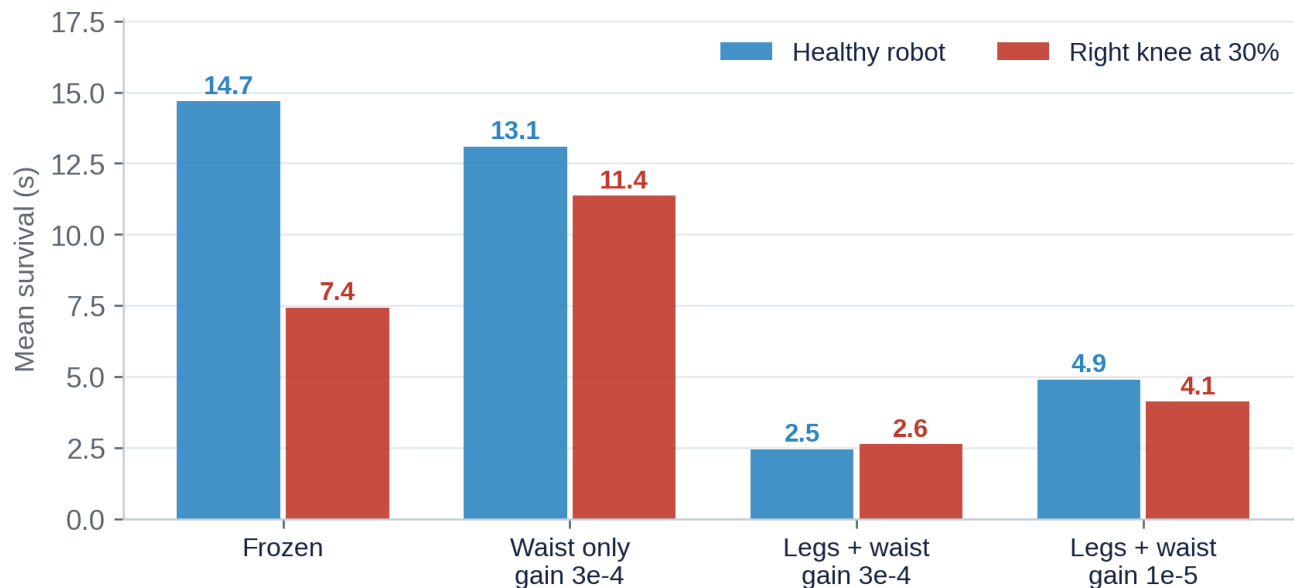
Humanoid policies can learn manipulation and locomotion that works.

Resilience is a different problem. It requires the robot to respond to changes in its own dynamics.

This summer produced the behaviors, the deployment path, and the adaptation framework that gets us there, with evidence that adaptation recovers a task the frozen policy loses.

Backup

The same law on a healthy robot spends margin the task needed



- **The tracking claim reproduces.** On the window every variant survived, leg tracking error goes from 14.82° frozen to 8.01° adapted. That is a 40% reduction, against 37% reported in the original MuJoCo testbed.
- **What does not transfer is the implication.** The published baseline falls at 1.94 s and never completes the motion, so adaptation could only add. Ours survives 14.68 s and completes the task on 6 of 6, and the same law spends that margin buying tracking accuracy the task did not need.
- **This is why the fault experiment matters.** A recovery controller applied to a healthy robot is outside the domain of the method. The comparison that means something is the one under the fault.

Controls: the numpy export matches the torch checkpoint to 4.2e-6 max action difference over 200 steps, and the baseline is that same numpy policy at gain zero, so the only difference between arms is the adaptation. A 1e-6 action perturbation moves leg error by 1.7° over 2.4 s, so every comparison is multi-seed.

Inverse-inertia weighting (gx level 2) is worse in both regimes. That reproduces the original finding: the weighting aims the correction at the lightest joints, the wrists and head, whose tracking errors are irrelevant to the task.

Diagnosing a sim-to-real failure with matched simulation

Symptom

All four takes of v33 abort on roll between 27% and 48% of the motion, always in the same direction.

First hypothesis, wrong

Compared against the only rollout on disk, which was a different policy. That pointed at a saturated ankle-roll command.

Matched baseline

Identified the exact v33 reference clip and ran the same policy, same clip, 3 seeds. Simulation clips that joint on 92% of ticks and still succeeds, so saturation is learned behavior, not the gap.

Actual cause

The deployment fed the torso IMU into `base_ang_vel`, which holosoma builds from the pelvis. The torso sits three waist joints above it, so this substitutes both a different body and a different frame.

Confirmation

Reproducing that substitution in simulation drops the policy from 100% success and 9.8 s to 0% and 1.6 s, falling in the same 1.1 to 1.9 s window as the hardware.

Fix

A pelvis estimator composes the torso IMU with the measured waist joints. Removes 90.8% of the observation error and restores 100% success.

Method that found it: a single tracking-error number conflates command saturation, servo error and task error. Separating the three is what localized this, and reproducing the defect in simulation is what proved it.

Configuration and reproduction

TRAINING		ADAPTATION	
Framework	holosoma whole-body tracking on IsaacLab / Isaac Sim	Law	ACC 2026 causality-guided layer adaptation, W_0 leak
Algorithm	PPO, $\lambda = 0.95$, $\gamma = 0.99$, 4096 envs	Adapted	layer index 2, $256 \times 128 = 32,768$ weights
Rates	200 Hz physics, 50 Hz control	Gain	$\Gamma = 3e-4$, leak 0.01, engage step 0
Actor	blind MLP 512-256-128 (ELU), 31 joint targets	Mask	waist yaw, pitch, roll
Critic	privileged, adds base linear velocity and object state	Fault	right_knee_joint PD stiffness $\times 0.3$
Pickup DR	box mass 0.4 to 1.6 kg, friction, PD gain, action delay, encoder noise, pushes	Protocol	6 seeds, 734 steps, no DR, no observation noise
Motion	OmniRetarget sub3_largebox_003, CMU crawl_111_03	Baseline	same numpy policy at gain 0

Full reproduction instructions, per-seed results and the hardware protocol are in adaptation/README.md and SETUP_ISAAC.md.

Repository: Agibot-humanoid · box_pickup/ adaptation/ agibot_control_functions/